

MANAR SALAH

Flutter Developer

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SUMMARY

Flutter Developer with 2+ years of experience building and delivering real-world mobile applications. Focused on turning product ideas into reliable and scalable applications, with hands-on experience in application architecture, backend integration, AI-powered features, and complex product workflows. Looking to join a strong engineering team to contribute to meaningful products, take ownership of challenging technical problems, and continue growing as a software engineer.

PROJECT EXPERIENCE

ZooVerse — Multi-Role Zoo Management & Visitor Platform *(Case Study)* Jun. 2026

Technologies: Flutter, Dart, Firebase, Cloud Firestore, Supabase, BLoC/Cubit, Clean Architecture, GoRouter, Dependency Injection, Geofencing, OpenStreetMap, Google Maps, Paymob Payment Gateway, Real-time Systems, Localization, Responsive Web

- Delivered **3 integrated applications** — a Visitor mobile app, an Employee mobile app with role-based workflows, and an Admin web dashboard — as a single zoo management ecosystem on a shared backend.
- Delivered core visitor journeys covering animal discovery, **Google Maps navigation**, and QR-based ticket entry with **Paymob payment integration**.
- Engineered real-time animal assignment using **Firestore transactions**, preventing conflicting claims between multiple keepers.
- Established a full **emergency-response lifecycle** with priority levels, evidence capture, and status tracking from Pending to Resolved.
- Engineered employee workflows for ticket validation, **duplicate-entry prevention**, and geofenced attendance tracking.
- Implemented animal-care reporting with feeding records, evidence uploads, and **critical-condition escalation** to administrators.

Graduation Assistant — AI-Powered Graduation Project Workspace Mar. 2026

Technologies: Flutter, Dart, BLoC, Clean Architecture, Gemini AI, Supabase, Hive

- Developed an **AI-powered workspace** that turns raw project ideas into structured deliverables — SRS, ERD, use cases, and risk analysis.
- Architected a **local-first data layer** with Hive, backed by bidirectional Supabase synchronization and conflict resolution.
- Integrated **Google Gemini** with prompt sanitization, structured JSON parsing, and rate-limited retry handling.
- Built a custom **2D diagram renderer** using Flutter Canvas to generate interactive ERD and Use Case diagrams with Bézier curve routing — no third-party library.
- Secured data ownership end-to-end with Supabase Auth, PostgreSQL, and **Row-Level Security (RLS)**.

Tabibi — Multi-Role Healthcare Management Platform *(Case Study)* Nov. 2025

Technologies: Flutter, Dart, Firebase, Cloud Firestore, BLoC/Cubit, Clean Architecture, GoRouter, Dependency Injection, Firebase Notifications, PDF Generation, Localization, Responsive Web, Custom Design System

- Delivered **3 integrated applications** — a Patient app, a Doctor app, and an Admin web dashboard — as a single healthcare platform on a shared real-time backend.
- Implemented the full **appointment lifecycle**, from booking and availability to cancellation and follow-up scheduling.
- Enabled **real-time doctor-patient messaging** with text, files, voice notes, and read receipts.
- Produced **digital clinical records** with diagnosis, prescriptions, and PDF medical reports.
- Implemented doctor access control with **admin-generated invitation codes** and role-based permissions.
- Established shared architecture across all 3 apps using **Clean Architecture** and reusable design-system components.

Ishara — Arabic Sign Language AI *(Case Study · GitHub)* Aug. 2025

Technologies: Flutter, Dart, MediaPipe, TensorFlow Lite, On-Device ML

- Built an **offline-first** Arabic Sign Language app with bidirectional translation: Sign → Text and Text → Sign, running entirely on-device with no cloud dependency.
- Achieved recognition support for **31 Arabic sign classes** through a real-time computer vision pipeline using MediaPipe hand landmarks and a custom TFLite classifier.

- Implemented **temporal prediction stabilization** to smooth frame-by-frame noise and increase recognition consistency in live video input.

EDUCATION

Obour Institutes — Bachelor of Computer Science

October 2022 – June 2026 | Cairo, Egypt | Grade: Good

TECHNICAL SKILLS

Mobile Development: Flutter, Dart, Android, Responsive UI

Architecture & State Management: Clean Architecture, BLoC/Cubit, Dependency Injection, Repository Pattern

Backend & APIs: Firebase, Supabase, PostgreSQL, REST APIs, Row-Level Security (RLS)

AI & Machine Learning: Google Gemini API, TensorFlow Lite, MediaPipe, Computer Vision

Data & Synchronization: Hive, Offline-First Architecture, Cloud Synchronization, Conflict Resolution

Tools & Testing: Unit Testing, Git, GitHub, Debugging

UI & Internationalization: Material 3, Arabic/English Localization, RTL Support

Soft Skills: Problem-Solving, Attention to Detail, Adaptability, Time Management, Self-Directed Learning, Ownership

LANGUAGES

Arabic: Native | **English:** Fluent